A Comprehensive Study of Privacy Preservation of Machine Learning Techniques by Using Blockchain

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Author Declaration

I hereby declare that this thesis is my own work and that I have clearly acknowledged all sources used in its creation. No portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification at this or any other university or institute of learning.

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Abstract

The growing digital world has made it more and more important to have secure and privacy aware fraud detection systems, as machine learning models trained on sensitive financial data remain vulnerable to inference and data leakage attacks. Differential Privacy offers protection against such risks, yet its impact on model performance in fraud scenarios requires further investigation. Meanwhile, blockchain technology, known for its decentralized and tamper proof structure, provides strong guarantees for transparency and accuracy.

This study proposes a privacy preserving fraud detection framework, uses two types of models: a Multi-Layer Perceptron (MLP) and a Random Forest (RF). The MLP is improved using Differential Privacy (DP-SGD) and incorporates a blockchain inspired logging layer, while the Random Forest serves as a classical ensemble learning baseline for benchmarking. Using a 2023 credit card fraud dataset, aim of the study is to evaluate the privacy utility trade-off and examine how blockchain logging improves transparency while keeping sensitive data protected. .

Keywords: Differential Privacy, Blockchain, Privacy Preservation, Erp, Machine Learning

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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CCS	Computer and Communications Security
DP	Differential Privacy
DP-MLP	Differentially Private Multi-Layer Perceptron
DP-SGD	Differentially Private Stochastic Gradient Descent
ERP	Enterprise Resource Planning
F1	F1 Score
FL	Federated Learning
FN	False Negative
FP	False Positive
GDPR	General Data Protection Regulation
HE	Homomorphic Encryption
IoT	Internet of Things
ML	Machine Learning
MLP	Multi-Layer Perceptron
PPML	Privacy-Preserving Machine Learning
RF	Random Forest
ROC	Receiver Operating Characteristic
SGD	Stochastic Gradient Descent
SMPC	Secure Multi-Party Computation
TN	True Negative
TP	True Positive

Chapter 1 Introduction

The increasing use of machine learning in industries has led to the collection and processing of large amounts of sensitive data. Most of the practical applications, training datasets contain personal or confidential information, which raises significant privacy concerns. Recent researchs shows that machine learning models may unintentionally reveal information about its training data, creating risks that go beyond traditional data security measures Ye & Shokri (2025), Song & Mittal (2019).

Inference attacks show that an adversary can analyze model outputs to see if a particular record was included in the training set. Membership inference attacks particularly reveal that models may behave differently for training and non training samples, allowing attackers to extract sensitive membership information Ye & Shokri (2025), Song & Mittal (2019)

Differential Privacy offers a formal mechanism to solve this problem by limiting the influence of each individual record on the final output of the model. It lowers the risk of sensitive information leakage while retaining its predictive value by injecting calibrated noise during computation Hassan et al. (2020).Concurrently, blockchain technology provides record keeping that is decentralized and protected against modification, allowing transparent and verifiable logging without relying on a trusted third party Chen et al. (2020). Adding these approaches together makes it possible to design systems that support both confidentiality and integrity.

Motivated by these challenges, this thesis proposes a fraud detection framework that integrates privacy preserving learning with blockchain based logging. Aim is to protect sensitive information while maintaining reliable detection performance and ensuring that the training process remains auditable.

1.1 Background

Machine learning models learn statistical patterns from historical data and use them to make predictions on unseen instances. When trained on sensitive datasets, these models may memorize specific details about individual records. These memories rise the privacy risks, since attackers can laverage model behavior to infer private

information.

Membership inference has become a standard method for evaluating these risks. Aim to determine whether a specific record was included in a model's training dataset by exploiting differences in model behavior between training and unseen samples. It measures whether an adversary can distinguish between records that were used for training and those that were not. Systematic evaluations show that privacy risk is not uniform across all samples and that some records may be more vulnerable than others Song & Mittal (2019). This highlights the need for explicit privacy protection during model training.

Differential Privacy solves this problem by ensuring that the presence or absence of a single record does not have an impact on the learned model. Noise is introduced into the learning process to reduce information leakage and ensure quantifiable privacy guarantees Hassan et al. (2020). This process may protect and effect accuracy, but it is a principled method of processing sensitive data

In addition to confidentiality, many distributed systems need integrity and traceability as well. Blockchain systems have an immutable ledger with records that cannot change without discovery. These properties support secure operation logging and auditing and, accordingly, are appropriate for encoding workflows in machine learning technology Chen et al. (2020). So, all together, Differential Privacy and blockchain can be complementary mechanisms that protect both data privacy and system integrity.

1.2 Problem Statement

Machine learning models are commonly used for fraud detection because they can capture complex patterns in financial transaction data. However, these models are usually trained on highly sensitive and personally identifiable information, which creates privacy risks. Recent research has shown that models may unintentionally leak sensitive information through attacks such as membership inference, model inversion, and gradient-based reconstruction Shokri et al. (2017), Zhu et al. (2019). This problem is especially important in financial areas, where confidentiality is mandatory and regulated by frameworks such as GDPR.

Although Differential Privacy (DP) provides a mathematical way to limit the contribution of individual samples and reduce privacy leakage, it usually reduces performance by adding noise during training. As a result, it remains unclear how well DP-based models can maintain accuracy in real fraud detection settings, especially when using neural networks such as Multi-Layer Perceptrons (MLPs).

At the same time, blockchain technologies provide immutability, tamper proof,

and transparent auditing. Many studies use blockchain for secure logging, but its use in privacy preserving machine learning is still limited. Despite progress in PPML techniques, the combined use of Differential Privacy and blockchain inspired logging within a practical fraud detection model has not been sufficiently investigated, particularly on modern datasets such as the 2023 credit card fraud dataset used in this thesis.

1.3 Thesis Objectives

The goal of this study is to develop and evaluate a privacy conscious and reliable machine learning architecture for sensitive data settings. Then, fraud detection is used as an illustrative application domain to validate the proposed method. Such as:

• Create a machine learning detection framework to detect anomalous or fraudulent patterns. • Use Differential Privacy in training to minimize leaking of information from sensitive data • Investigate the effects of privacy protection on predictive performance and assess the trade-off between privacy and model usefulness • integrate a blockchain logging mechanism to ensure transparency, integrity, and tamper resistant record keeping of the training and evaluation processes • evaluate the applicability of the proposed framework in enterprise and real world operational contexts.

1.4 Research Questions

This thesis addresses this challenge by investigating how privacy preserving learning methods and secure logging systems can be added to a company's fraud detection setups without taking away from their ability to make accurate predictions. In particular, this study following research questions such as:

RQ1: How can machine learning models be trained to detect fraud while keeping personal and transaction details safe using Differential Privacy?

RQ2: What effect does Differential Privacy have on how well a model performs, and how does privacy and utility trade-offs affect fraud detection effectiveness?

RQ3: How can a blockchain logging mechanism help make sure that transparency, integrity, and immutable logs are present during the model training and evaluation process?

RQ4: Can privacy preservation and data integrity guarantees be incorporated without significantly degrading classification performance?

RQ5: How can privacy aware and trackable framework such as blochchain can

be added to enterprise resource planning (ERP) systems for real life use?

1.5 Thesis Thesis Organization

Thesis is organized as follows,

Chapter 1 is the part that introduces the research problem, gives the background, and presents the motivation for securing the data in machine learning; fraud detection, with research questions situated. Chapter 2 provides relevant literature, including fraud detection models, privacy risks in machine learning, Differential Privacy, blockchain technology, and existing PPML approaches. In Chapter 3, the proposed methodology is described, detailing the dataset, pre processing procedures, the MLP model, the Differential Privacy mechanism, and the blockchain logging layer. Chapter 4 presents the results and discussions, where the findings are analyzed in relation to the research objectives. Chapter 5, discusses extended application scenarios, particularly the applicability of the framework within ERP systems Finally, Chapter 6 concludes the conclusion, summarizing the key findings, contributions, limitations, and directions for future research.

Chapter 2 Literature Review

This chapter summarizes the current literature on fraud detection, techniques used for machine learning, privacy preserving methods, and blockchain based systems. It describes prior methods, their limitations, and highlights the research gap that motivates the proposed framework. The chapter provides theoretical foundations for the methods adopted in this study

2.1 Machine Learning for Fraud Detection

Machine learning is a subset of artificial intelligence (AI) that focuses on developing algorithms capable of learning patterns from training data and subsequently making accurate inferences on unseen data Bishop (2006). These techniques have become a central component of modern fraud detection systems due to their ability to analyze largescale transactional data and automatically identify suspicious behavioral patterns. Compared to traditional rule-based approaches, learning-based models adapt more effectively to evolving fraud strategies and provide improved generalization performance. Comprehensive surveys report that data mining and machine learning methods consistently achieve higher detection rates in financial fraud scenarios Ngai et al. (2011), Phua et al. (2010).

The training of such models generally follows two primary paradigms: supervised and unsupervised learning Bishop (2006), Han et al. (2012). In supervised learning, historical transactions are labeled as legitimate or fraudulent and used to train classifiers that learn discriminative patterns between classes. In contrast, unsupervised learning operates without labeled data and aims to discover inherent structures or anomalies within the dataset, which is particularly useful for identifying previously unseen or emerging fraud patterns.

Supervised learning remains widely adopted in this domain, where labeled historical data enable the development of accurate predictive models. Ensemble-based methods such as Random Forest are frequently employed as strong baselines because of their robustness to overfitting and their interpretability through feature importance analysis Breiman (2001). Neural network models, particularly Multi-

Layer Perceptrons, are capable of modeling nonlinear feature interactions and have demonstrated competitive performance in structured transaction datasets Goodfellow et al. (2016).

In addition, anomaly detection techniques have been explored to detect suspicious behaviors without relying on labeled data Chandola et al. (2009). Nevertheless, studies emphasize that automated detection systems often benefit from complementary human expertise when handling complex fraud cases Bolton & Hand (2002).

2.2 Multi-Layer Perceptron Models

The Multi-Layer Perceptron (MLP) is one of the most widely used feedforward neural network architectures for classification tasks. An MLP consists of multiple fully connected layers and nonlinear activation functions, enabling the model to learn complex decision boundaries from input features Goodfellow et al. (2016).

Due to their ability to capture nonlinear relationships among attributes, MLPs are particularly suitable for fraud detection problems where transaction patterns are often subtle and high dimensional. Compared to shallow models, neural networks can learn richer representations of behavioral signals, which may improve detection performance in highly imbalanced financial datasets. For this reason, MLP models are frequently adopted as primary classifiers in modern fraud detection studies and serve as the main learning architecture in this thesis.

2.3 Privacy Risks in Machine Learning

Training advanced machine learning models needs a huge amount of data, which in many cases can contain personal and sensitive data. A recent thread of researches has shown that machine learning (ML) models memorize sensitive information of training data, indicating serious privacy risks. A machine learning model might reveal different types of information about the data records in its training data. On the one hand, the statistical patterns about the training data are indeed what we expect to learn from the model (in order to successfully perform various downstream tasks). On the other hand, certain information could be very specific to an individual, and revealing it might result in a serious data privacy violation. So, the "information" that can be learned from a model may be of different types, among them only some types can be considered private. It is, therefore, very important to have a clear definition of data privacy risks as a specific type of information leakage.

We need to identify the type of information that is very specific to any given individual data record z in the training set, when cannot be learned from and be

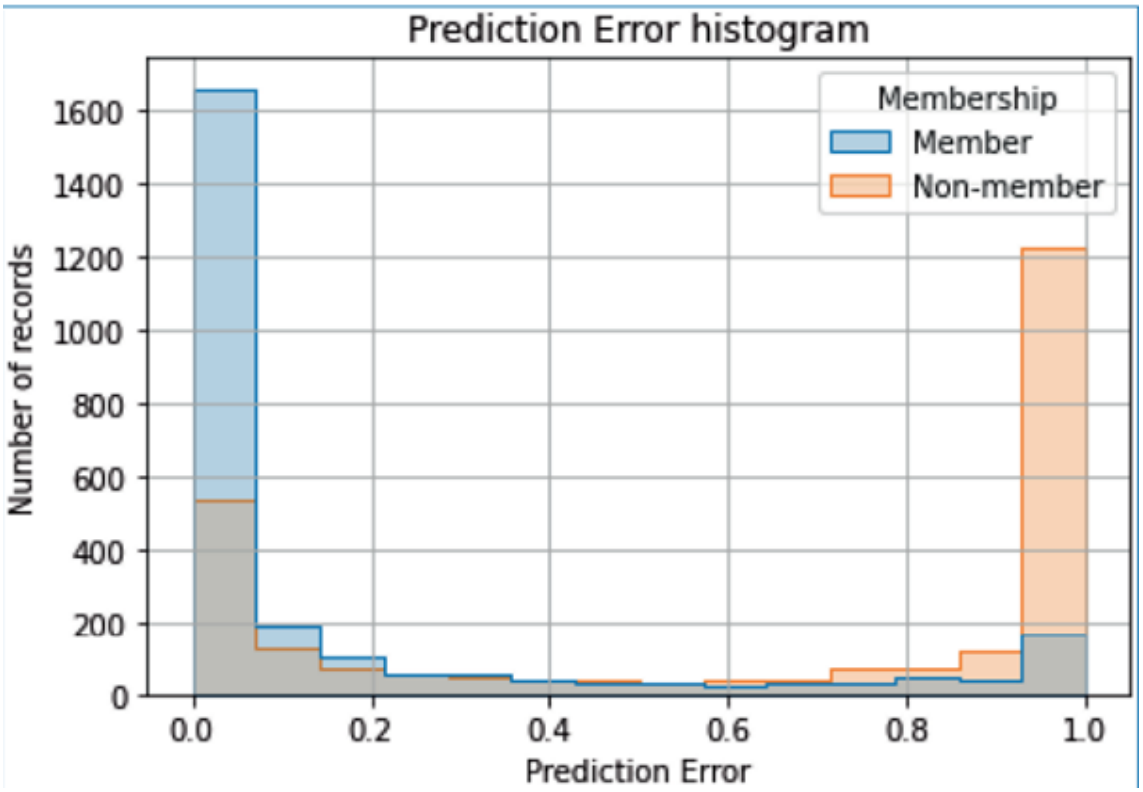

Figure 2.1: The prediction error of a target model

associated with the training data as a whole when excluding z. One of the most fundamental forms of specific information that a model might leak about its training data is their membership, i.e. to reveal that a data record is indeed used for training the model. Modeling this type of information leakage is crucial for defining privacy risks, as it could be considered as the foundation of privacy attacks. In fact, membership inference attacks against mlp often serve as the building block for other empirical inference attacks, such as reconstruction attacks and attribute inference attacks. Therefore, in this chapter, we focus on the leakage of membership information as a way to quantitatively reason about privacy violation in machine learning. Privacy risk in terms of information leakage essentially boils down to revealing uniquely identifying patterns of training data associated with a model. These patterns make training set members distinguishable from non-member instances (population data), thus revealing the sensitive membership information if such patterns are observable by the adversary.

MLP trained on sensitive data may unintentionally leak information about individual records. This risk arises because models can memorize characteristics of training samples, especially when overfitting occurs. As a result, adversaries may exploit model outputs to infer whether a specific data record was used during training, a threat known as a membership inference attack.

Figure 2.3 shows distribution of prediction errors for member and non member

samples. Member samples used during training exhibit lower prediction error than non-member samples, indicating memorization effects in the model. This separation enables membership inference attacks by allowing adversaries to distinguish training data from unseen records, revealing privacy vulnerabilities in standard MLP

Such privacy risks are particularly critical in domains like fraud detection and financial analytics, where models are trained on highly sensitive transactional data. These vulnerabilities highlight the need for privacy-preserving mechanisms that limit the influence of individual data records during model training, motivating the use of Differential Privacy in this study.

This behavior demonstrates a fundamental privacy risk in machine learning models trained on sensitive data. To mitigate such membership inference vulnerabilities, privacy preserving techniques such as Differential Privacy are required, motivating the approach adopted in this study.

2.4 Differential Privacy and DP-SGD

Differential Privacy provides a mathematical framework for protecting individual data records during computation. It guarantees that the inclusion or exclusion of a single training sample does not significantly affect the model output, thereby limiting information leakage.

The theoretical foundations of Differential Privacy were established by Dwork et al., who proposed adding carefully calibrated noise to bound query sensitivity Dwork et al. (2006). Extending this concept to deep learning, Abadi et al. introduced Differentially Private Stochastic Gradient Descent (DP-SGD), which clips per-sample gradients and injects noise before parameter updates Abadi et al. (2016). This mechanism restricts the influence of individual samples and provides provable privacy guarantees against inference attacks.

Although noise injection in differentially private stochastic gradient descent (DP-SGD) introduces a trade-off between privacy and predictive performance, prior empirical studies show that models trained with moderate privacy budgets can still achieve competitive accuracy when properly tuned Bagdasaryan et al. (2019), De et al. (2022). Therefore, DP-SGD has become a practical technique for training neural networks on sensitive financial datasets.

2.5 Blockchain Technology and Logging

Beyond privacy protection, ensuring the integrity and transparency of the training process is also important, especially in regulated financial environments that require

auditability. Blockchain technology has been explored as a solution for secure record keeping that cannot be modified.

Blockchain systems maintain an immutable and append only ledger, where records are cryptographically linked to prevent unauthorized modification. Originally proposed by Nakamoto Nakamoto (2008), this architecture provides strong guarantees of integrity and traceability. Surveys on blockchain technologies highlight their usefulness in secure logging and auditing applications Zheng et al. (2017). Furthermore, blockchain-based logging mechanisms have been shown to protect event histories against deletion and tampering, improving the reliability of forensic analysis Cai et al. (2018).

Figure 2.1 illustrates the blockchain system architecture, which consists of five layers, each with distinct responsibilities to enable decentralized, secure data storage and perform various transactions. The innate modular design allows different components to be further improved in terms of efficiency and performance. These layers facilitate cryptography, networking, consensus protocols, and other functions needed for transparent ledgers without centralized control.Ahmed & Alabi (2024)

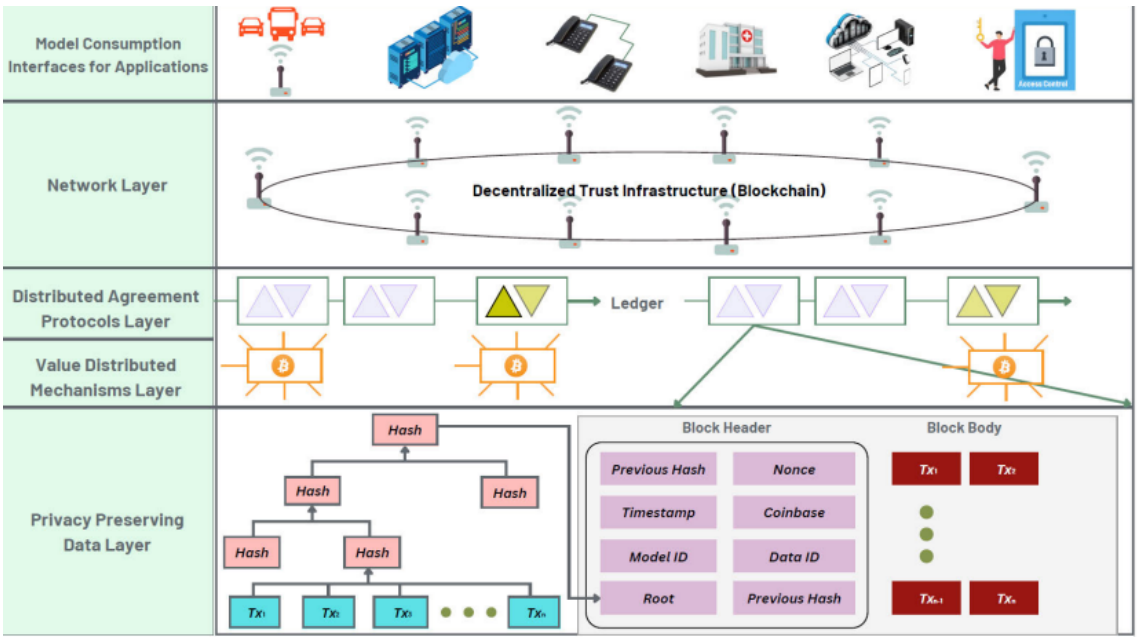

Figure 2.2: Blockchain system architecture

The functionalities of the five layers are described as follows:

• **Model Consumption** Interfaces for Application: It is also called the application layer, which enables user interaction with the applications deployed on the blockchain network.

• **Network Layer:** It facilitates node communication and data transmission.

• **Consensus Layer:** It maintains that all network nodes agree on the current blockchain state using consensus algorithms.

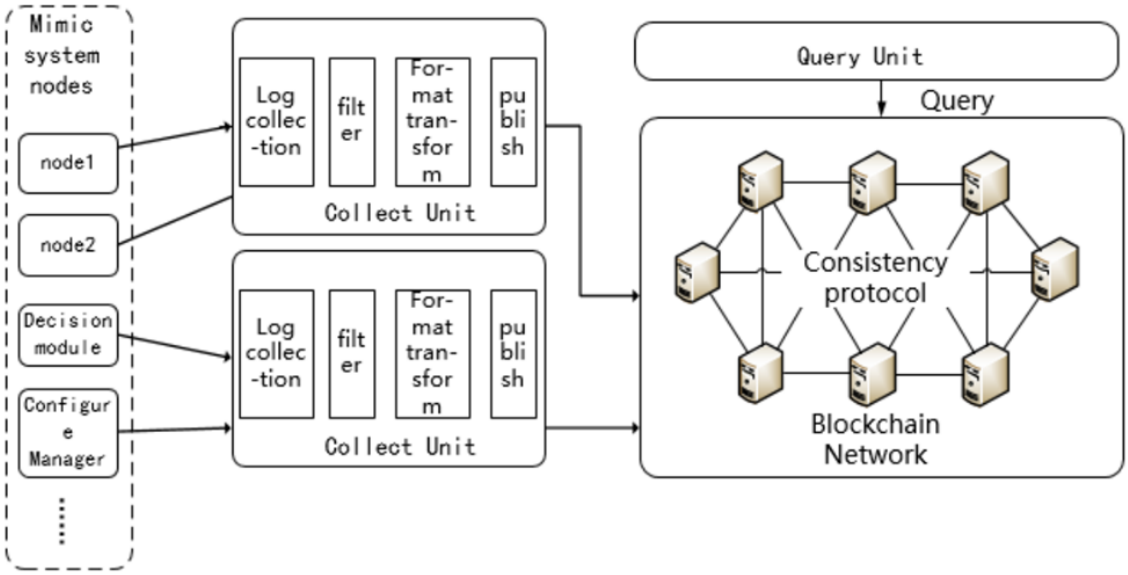

Figure 2.3: Architecture of Log system with Blockchain inspired

• **Value Distribution** Mechanism Layer: It is also called the incentive layer, which rewards nodes for contributing to the network's performance and security.

• **Data Layer:** It maintains data security, integrity, and immutability on the blockchain

However, deploying a full blockchain infrastructure may introduce significant computational overhead. Consequently, lightweight blockchain-inspired logging approaches are often preferred for machine learning workflows, providing integrity guarantees while maintaining system efficiency.

Ahmed & Alabi (2024)

Figure 2.3 is a Blockchain inspired logging architecture. System events are collected, filtered, and transformed before being published to a distributed blockchain network that ensures integrity and consistency. The query unit enables secure auditing and verification of recorded logs.

2.6 Existing Benchmarks

Benchmarking plays an important role in evaluating fraud detection systems. Public credit card transaction datasets are commonly used to compare machine learning models under highly imbalanced class distributions. In such settings, performance metrics such as precision, recall, and F1-score are preferred over simple accuracy, which may be misleading when fraudulent cases are rare.

Previous studies compare classical machine learning models, ensemble approaches, and neural networks on these datasets to assess detection effectiveness Dalal & Abdul (2017). With the growing interest in privacy preserving learning, recent benchmarks

also evaluate the trade-off between privacy strength and model utility when Differential Privacy mechanisms are applied. These evaluations provide a foundation for assessing both predictive performance and privacy guarantees in fraud detection tasks.

2.7 Related Work

Existing research has extensively studied fraud detection, privacy-preserving learning, and secure logging independently. Fraud detection studies primarily focus on improving classification performance using advanced machine learning architectures. Privacy research concentrates on mitigating inference attacks through techniques such as Differential Privacy. Meanwhile, blockchain based approaches emphasize integrity, transparency, and protected logs of recorded information.

However, limited work integrates these directions within a unified framework. In particular, the combination of differentially private neural network training with blockchain logging for fraud detection remains underexplored. Addressing this gap, this thesis proposes a comprehensive framework that jointly ensures data confidentiality through Differential Privacy and process integrity through tamper-resistant logging while maintaining competitive detection performance.

Chapter 3 Methodology

This chapter presents the construction of the secure fraud detection framework proposed in this paper along with the experimental procedures. The dataset characteristics, preprocessing steps, baseline models, differential privacy implementation, blockchain logging mechanism, and the evaluation strategy. And lastly, the overall framework and system design.

Section 3.1 presents the overall framework and system design. Characteristics and the dataset are described in Section 3.2. 3.3 is where Data preprocessing is outlined The following sections report the baseline models (e.g. Multi-Layer Perceptron and Random Forest) in Section 3.4. In Section 3.5, Differential Privacy implementation with DP-SGD is described. Section 3.6 presents the blockchain inspired logging mechanism. Section 3.7 details the benchmarking strategy and Section 3.8 describes the evaluation criteria and experimental setup.

3.1 Proposed Framework

This study proposes a comprehensive fraud detection, protected data framework that integrates Differential Privacy with a blockchain-inspired logging mechanism. The motivation behind this framework is to address two critical challenges commonly encountered in financial machine learning applications: the protection of sensitive transaction data and the transparency of the model training process. The framework is built as a series of separate steps that work one after another, so each part can function on its own but still helps achieve the main goal of the whole system. First, raw credit card transaction data are collected and prepared for analysis. These data go through preprocessing steps to make sure they are stable and work well with neural network learning methods. After preparing the data, a basic Multi-Layer Perceptron (MLP) model is trained using usual methods for improving performance. This baseline model serves as a reference point, providing an upper-bound estimate of model performance in the absence of any privacy constraints.It's important to set this reference so can properly assess how the privacy preserving methods later impacts in the framework to enforce formal privacy guarantees,

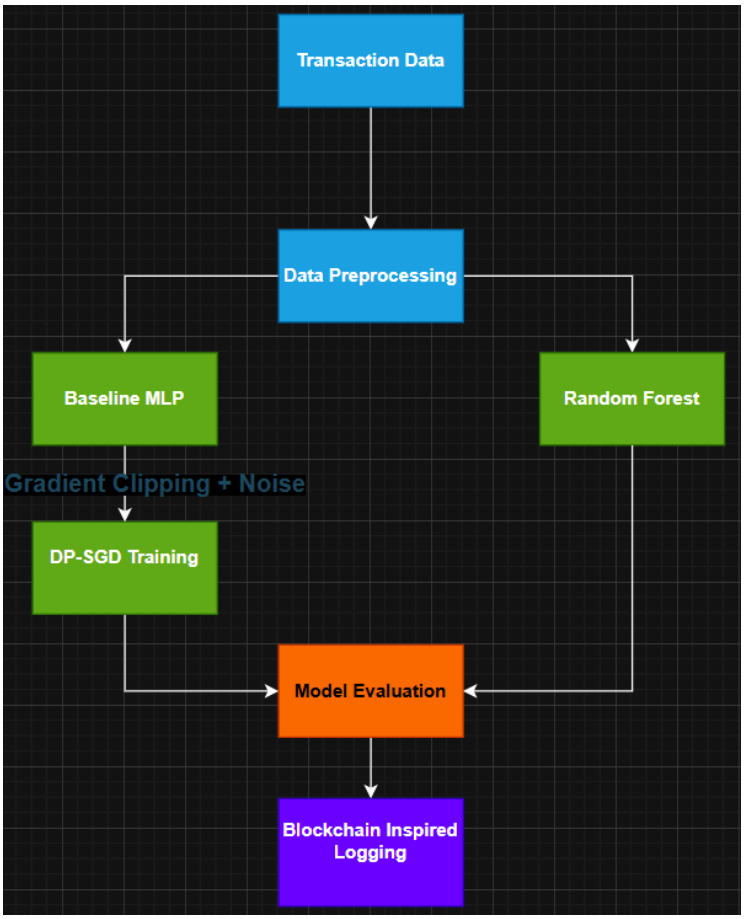

Figure 3.1: Proposed Framework

the training process is subsequently modified using Differentially Private Stochastic Gradient Descent (DP-SGD). By using gradient clipping and adding some noise, DP-SGD makes sure that each training example doesn't have too much effect on the model's learned parameters.This change allows the framework to reduce the chances of privacy information being exposed through attacks that use inference.In parallel with the privacy-aware training process, a blockchain-inspired logging component is added into the framework. This component records critical training metadata, including loss values, accuracy metrics, privacy parameters, and model hashes, in an immutable structure.Instead of using a complete blockchain system, a simpler design that is inspired by blockchain is used. This helps keep things clear and trackable, while also avoiding heavy use of computer resources. Overall, this end-to-end framework enables a systematic analysis of the privacy–utility trade-off in fraud detection while maintaining verifiable and tamper-resistant training records

Figure 3.1 illustrates the overall workflow of the proposed fraud detection framework. After data pre processing, multiple models including a baseline MLP, a differentially private DP-MLP, and a Random Forest benchmark are trained and evaluated. The training process is monitored through a blockchain inspired logging

mechanism to ensure integrity and transparency.

3.2 Dataset Description

The experimental evaluation of the proposed framework uses a credit card transaction dataset released in 2023, founded in Kaggle. The dataset contains 568,630 transactions. Each transaction has 31 features. To simplify the comparison between models and to avoid extreme class imbalance, a balanced subset of the dataset was constructed. Although this balanced distribution does not reflect real world fraud rates, it was intentionally selected to enable controlled comparison between baseline and privacy-preserving models. This setup allows the impact of Differential Privacy and the logging mechanism to be evaluated more clearly without being dominated by extreme class imbalance. In this subset, the number of normal and fraudulent transactions is equal (284,315 each), resulting in a 50% fraud rate during training and evaluation. Which in real life situation that is rate is close to impossible. But this setup allows a clearer analysis of the effects of Differential Privacy and logging mechanisms on model behavior. These include 28 anonymized numerical features from sensitive transaction data, the transaction amount, and a label indicating whether the transaction is fraudulent. The sensitive information is anonymized so that individuals cannot be identified, while the data can still be used for fraud detection. Dataset is different from most real life fraud detection, it doesn't have a big imbalance between normal and fraudulent transactions. Because of real fraud datasets are rarely shared publicly due to privacy, security, and legal concerns. Instead, it has a fair mix of both types of transactions. Overall, the dataset provides a realistic and conscious privacy basis for testing fraud detection models in supervised learning, making it ideal for analyzing how Differential Privacy affects model performance. The transformed feature space make it easier to clear separation of the two classes, which may lead to unusually high classification performance compared to typical real world fraud detection scenarios. As a result, most machine learning models are likely to show very high accuracy and AUC scores. So, the main goal of this study isn't just to get the best predictions, but but rather to evaluate how PPML and blockchain based integrity mechanisms influence model behavior under controlled experimental conditions.

3.3 Data Preprocessing

Before starting the training of the model, the dataset went through several steps to prepare it properly. These steps helped make the data stable and ready for

use with neural networks. Since all the data in the dataset is made up of numbers, there was no need to use any methods to convert categorical data into numbers. The feature set was kept as it was, because these features were already scaled and modify during the anonymization process. In contrast, the transaction amount had a large range of numbers, so it was scaled to make sure it didn't have too much influence on the optimization process during training.After scaling the features, the dataset was divided into training and testing subsets. An 80/20 split was applied, where 80% of the data was used for training and 20% for testing. A fixed random seed (random_state = 42) was used to ensure reproducibility, and stratified sampling was employed to preserve the class distribution in both sets. Steps were taken to keep the balance of classes the same in both groups to ensure a fair and consistent evaluation of model performance.No techniques like resampling or adjusting class balances were used during the preprocessing stage. This decision lets the experiment focus on how Differential Privacy methods affect results, instead of being influenced by how imbalance is handled.

3.4 Baseline Models

To establish reliable performance benchmarks, two baseline models are used: on based on neural network method called Multi-Layer Perceptron (MLP) and a traditional ensemble method, Random Forest (RF). Using both models allows for a fair comparison between deep learning and traditional machine learning approaches for fraud detection on structured financial data.

3.4.1 Multi-Layer Perceptron (MLP)

A baseline Multi-Layer Perceptron (MLP) model was implemented to provide as a benchmark for fraud detection without using any methods to protect privacy. This baseline represents the standard training setting where transaction data is used as is, without adding any Differential Privacy or additional protection mechanism.

The model takes the numerical transaction features as input and consists of several fully connected layers. In our implementation, two hidden layers with 128 and 64 neurons were used. ReLU activation functions were applied between layers to capture non-linear relationships in the data. ReLU activation functions were applied between layers to capture non-linear relationships in the data.The output layer uses a sigmoid activation function to perform binary classification between legitimate and fraudulent transactions.

The model was trained using the Adam optimizer with a learning rate set to 0.001 and a batch size of 512.The goal was to use binary cross entropy loss as the

main measure to minimize.Training was performed for 7 epochs.No gradient clipping or noise injection was applied in this stage. So this way, baseline gives an idea of the best possible prediction results you can get before adding Differential Privacy.

3.4.2 Random Forest (RF)

Besides the neural network model, Random Forest classifier was used as a traditional machine learning baseline so can see the benchmarking between an neural network and traditional.Random Forest is a tree based ensemble method that combines several decision trees and is often used for solving classification tasks involving structured data.

In this study, the Random Forest model was trained using 200 trees with default parameters. Model learns decision rules from the transaction features and aggregates the predictions of individual trees to produce the final output. This approach provides a strong and stable benchmark for comparison with neural network-based models

Random Forest model is different from the DP-MLP because it doesn't have any privacy protections and is trained with the original data. The use of classical machine learning methods is only to evaluate how they perform compared to the proposed privacy-preserving neural network framework.

3.5 Differential Privacy Implementation

Differential Privacy was integrated into the training process to minimize the risk of sensitive information leakage from the learned model. In fraud detection systems, models are trained usinig very sensitive transaction data. Without proper protection, trained models may unintentionally reveal information about individual records through inference attacks. Therefore, privacy-preserving learning is necessary for ensuring secure and reliable implementation.

Differential Privacy provides a formal guarantee that the adding or removing of a single record does not significantly change the output of a learning algorithm. Following the original definition, a randomized mechanism $\mathcal{A}$ satisfies ε-Differential Privacy if, for any two neighboring datasets D and D' that differ in one record, and for any output set S Dwork et al. (2006),

$$P(\mathcal{A}(D) \in S) \leq e^{\varepsilon} P(\mathcal{A}(D') \in S).$$

In practice, pure ε-Differential Privacy may be too strict for machine learning tasks. Therefore, a relaxed version called (ε, δ)-Differential Privacy is commonly

used. This formulation allows a small probability of failure and provides a better balance between privacy and model performanceDwork et al. (2006):

$$P(\mathcal{A}(D) \in S) \leq e^{\varepsilon} P(\mathcal{A}(D') \in S) + \delta,$$

where ε controls the strength of privacy protection and δ represents a small failure probability. Smaller ε values provide stronger privacy but may reduce predictive accuracy.

To implement these guarantees, this study uses Differentially Private Stochastic Gradient Descent (DP-SGD). In this method, gradients are calculated for each training sample, clipped to a fixed norm to limit their influence, and Gaussian noise is added before updating the model parameters. This combination of gradient clipping and noise injection ensures that the training process meets the required privacy constraints. The noisy gradient update can be written as

$$\tilde{g} = g + \mathcal{N}(0, \sigma^2 C^2),$$

where C is the clipping norm and σ controls the noise scale.

The model that uses differential privacy is trained using the same conditions as the standard model. The only difference is the use of DP-SGD during optimization, which enables a fair and controlled comparison between private and non-private training.

3.6 Blockchain Inspired Logging

In addition to privacy preserving training, a blockchain inspired logging mechanism is introduced to enhance the transparency and reliability of the training process. Differential Privacy keeps sensitive information safe at the data level, but it doesn't guarantee that the training process is trackable or protected from later on modifications. The addition of a logging layer is added to record the main events during model training.

A full blockchain infrastructure was not adopted because it would introduce unnecessary computational overhead, latency, and deployment complexity for an experimental research setting. Therefore, a lightweight hash chained structure was preferred to provide integrity and tamper resistance with minimal system cost. The proposed solution follows the basic ideas of blockchain, such as hash-based chaining and immutability, without requiring distributed consensus or cryptocurrency components. This approach allows easy integration into a standard machine learning workflow.

During training, important metadata are recorded at each epoch. These records include training loss, evaluation metrics, privacy parameters (e.g., ε values), timestamps, and hashes of the model state. Each new record is linked to the previous one using a cryptographic hash. In this way If any previous entry is altered, the chain becomes invalid and the change can be detected immediately.

This structure creates an immutable audit trail of the training process. It improves experiment reproducibility and increases trust in the results. By combining Differential Privacy with blockchain inspired logging, the framework provides both data confidentiality and process transparency in a simple and practical way.

3.7 Benchmarking Approach

To assess the proposed framework, a benchmarking strategy was designed to compare different models under the same experimental conditions. The aim is to check how well predictions work and to look at how keeping privacy affects the results in a fair and repeatable manner.

Three model configurations are considered. The first is a Random Forest classifier, which serves as a traditional machine learning baseline.The second approach involves a standard Multi-Layer Perceptron (MLP) trained without any privacy protection. The third uses the same MLP architecture but is trained with Differentially Private Stochastic Gradient Descent. This setup allows comparisons between classical methods, neural networks, and privacy preserving learning within the same framework.

All models are trained and tested using identical data splits. The same steps are applied to each configuration to make sure consistency. For the MLP models, the architecture and training parameters are kept fixed so that any performance differences are caused only by the use of Differential Privacy.

This controlled design simplifies the process of interpreting the evaluation. Comparisons between Random Forest and MLP show how traditional methods differ from neural network approaches, while comparisons between the standard MLP and the DP-MLP highlight the privacy utility trade-off. Overall, this benchmarking process provides a clear and systematic assessment of the proposed system.

3.8 Evaluation Metrics

To evaluate the predictive performance of the models, several standard classification metrics were used. The same evaluation protocol was applied to the Random Forest, the baseline MLP, and the differentially private MLP to ensure fair comparison and

reproducibility.LaBelle AI (n.d.)

Accuracy measures the overall proportion of correctly classified transactions among all samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Although accuracy provides a general idea how is model performance, it may not fully reflect model quality in fraud detection tasks where the costs of wrong positive decision (false positive) or a wrong negative decision (false negative).

Precision evaluates how many predicted fraud cases are actually fraudulent:

$$Precision = \frac{TP}{TP + FP}$$

High precision indicates fewer false alarms.

Recall measures how many actual fraud cases are successfully detected:

$$Recall = \frac{TP}{TP + FN}$$

Recall is particularly important in fraud detection, since missing fraudulent transactions may lead to significant financial losses.LaBelle AI (n.d.)

To balance precision and recall, the **F1-score** was also computed:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

These metrics were selected because they provide complementary perspectives on classification performance and are widely used in fraud detection studies.

Finally, the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was calculated to assess the discrimination capability of each model across different decision thresholds.

Together, combined these metrics provide a comprehensive and reliable basis for benchmarking model performance and analyzing the impact of privacy-preserving mechanisms.

3.9 Experimental Setup

All experiments were conducted under controlled and reproducible conditions and consequently the reported results are directly comparable across model configurations. The dataset was partitioned and preprocessed in the same way throughout the study so that any difference we see in performance can be explained by either

model design or privacy related mechanisms

The training and testing data were divided into fixed training and test sets in a standard split ratio. The training part could exclusively be used for model learning, and the testing part was reserved for final performance evaluation. No information from the test set was used at the time of training and no data leakage occurred; it was used for unbiased assessment

For the neural network approaches, the same Multi-Layer Perceptron architecture and optimization settings were used for both the baseline and the differentially private models. Training was performed using mini batch optimization for a fixed number of epochs. Hyperparameters such as batch size, learning rate, and optimizer configuration were kept constant across experiments to isolate the effect of Differential Privacy.

The differentially private model employed Differentially Private Stochastic Gradient Descent with gradient clipping and Gaussian noise injection. Privacy related parameters, such as the clipping norm, noise multiplier, and privacy budget, were selected according to common practices in the literature. In addition, multiple privacy budgets were tested to analyze the trade-off between privacy strength and predictive performance.

For the Random Forest classifier, a fixed number of decision trees and consistent depth limits were defined to obtain stable and comparable results. Using standardized hyperparameters avoids excessive tuning and ensures a fair comparison with the neural network models. Lastly, all experiments were performed within the same software and hardware environment. This controlled setting ensures that any differences in performance are caused by the models themselves rather than external implementation factors.All neural network models were implemented using PyTorch, while scikit learn was used for preprocessing and the Random Forest baseline. Differential Privacy was integrated using the Opacus library.All experiments were conducted in the Google Colab environment using Python, which is easy to use and provides a reproducible platform for training and evaluation.

3.9.1 Implementation and Reproducibility

Implementation Details and Reproducibility

Implementation Details and Reproducibility

All experiments were conducted in the Google Colab environment using CPU execution (no GPU acceleration).

The implementation was developed in Python 3.12.12 using PyTorch 2.9.0 (CPU version), scikit-learn 1.6.1, and Opacus 1.5.4.

The baseline model was trained using the Adam optimizer with a learning rate

of 0.001, batch size of 512, binary cross-entropy loss, and 10 training epochs. The differentially private model used DP-SGD with gradient clipping and Opacus for privacy accounting.

To ensure reproducibility, the complete experimental notebook and source code are publicly available on GitHub.

https://github.com/RumeysaMercimek/Blockchain_ML

Chapter 4 Results and Discussion

Results and discussions chapter presents and analyzes the experimental results of all evaluated models. The predictive performance of the baseline, privacy-preserving, and classical approaches is compared using quantitative metrics and visualizations. The results are interpreted to examine the impact of Differential Privacy on model utility and to evaluate the effectiveness of the proposed framework. The outputs of the blockchain inspired logging mechanism are also discussed to demonstrate transparency and integrity.

4.1 Baseline MLP Results

To benchmark upper bound performance, the baseline Multi-Layer Perceptron (MLP) was trained without privacy-preserving constraints for fraud detection. In this setup, the typical learning condition is achieved, where the model has full access to the training data without noise injection or privacy related modifications.

The baseline MLP achieved very strong classification performance. As shown in Figure 4.1, the model correctly classified 56,814 legitimate and 56,710 fraudulent transactions. Only 49 false positives and 153 false negatives were observed. This corresponds to an accuracy of 99.82%, a precision of 99.91%, a recall of 99.73%, and an F1-score of 99.82%.

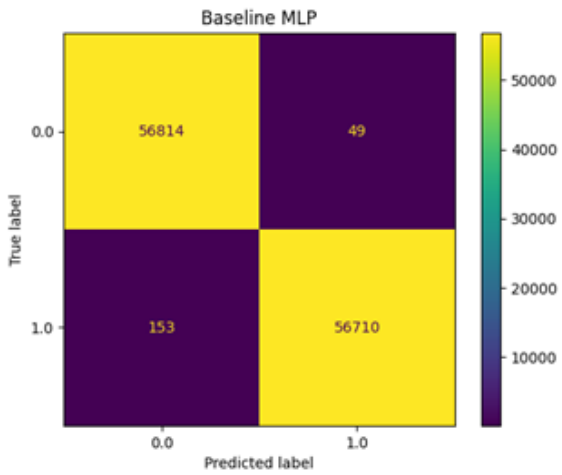

Figure 4.1: Confusion matrix of the baseline MLP model on the test set

These results imply that the anonymized features possess strong predictive information, which would help the model to perform efficiently in a relatively simple architecture. Thus, the baseline model acts as an upper bound benchmark and acts as a reference point for evaluating classical as well as privacy-preserving methods in the subsequent experiments.

4.2 Random Forest Results

To provide a classical machine learning baseline, a Random Forest classifier was trained using identical preprocessing steps and dataset splits.

As shown in Figure 4.2, the model correctly classified 56,805 legitimate and 56,229 fraudulent transactions, with 58 false positives and 634 false negatives. This corresponds to an accuracy of 99.33%, a precision of 99.90%, a recall of 98.88%, and an F1-score of 99.39%.

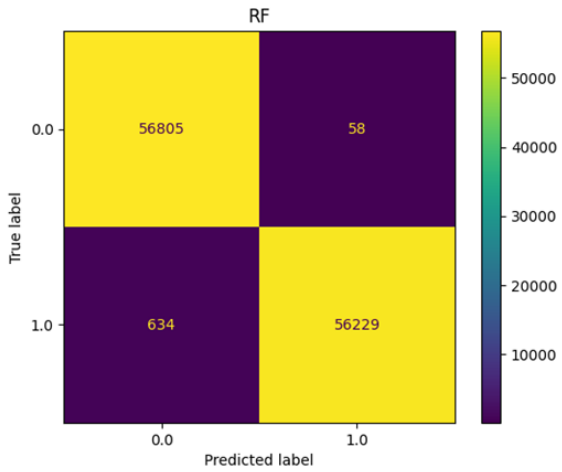

Figure 4.2: Confusion matrix of the Random Forest model on the test set

While the overall accuracy is already high, false negatives occur more than for the baseline MLP model. More fraudulent transactions are thus misclassified as legitimate, which may decrease detection effectiveness. These findings indicate that Random Forest is certainly a stable and interpretable baseline, but may have more difficulty at capturing complex non-linear relationships than neural network models. However, the classifier serves as a benchmark that is capable of comparison with both the baseline and privacy-preserving models.

4.3 DP-MLP Results

To protect sensitive training data, Differential Privacy was incorporated using the DP-SGD optimization procedure. The model was trained under a fixed privacy budget ($\epsilon = X$), introducing gradient clipping and calibrated Gaussian noise during optimization.

As shown in Figure 4.3, the model correctly classified 56,442 legitimate and 56,594 fraudulent transactions, with 421 false positives and 269 false negatives. This corresponds to an accuracy of 99.31%, a precision of 99.26%, a recall of 99.53%, and an F1-score of 99.39%.

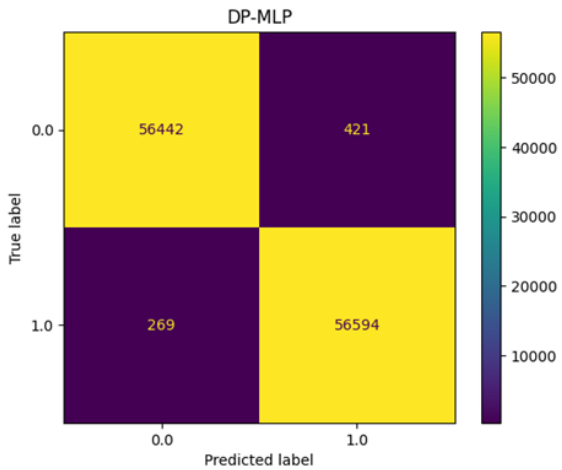

Figure 4.3: Confusion matrix of the DP-MLP model on the test set

Compared to the baseline models, the number of false positives increases due to the injected noise, which slightly reduces precision. However, the overall performance remains high, and most fraudulent transactions are still correctly detected. These results indicate that meaningful privacy protection can be achieved with only limited loss in predictive utility.

4.4 Privacy Utility Trade-Off

While Section 4.3 evaluates a single privacy configuration, additional experiments were conducted to analyze how varying privacy strengths influence predictive performance. The DP-MLP model was trained under multiple privacy budgets ($\epsilon \in \{0.5, 1, 2, 3, 5, 10\}$), and performance metrics were recorded for each configuration.

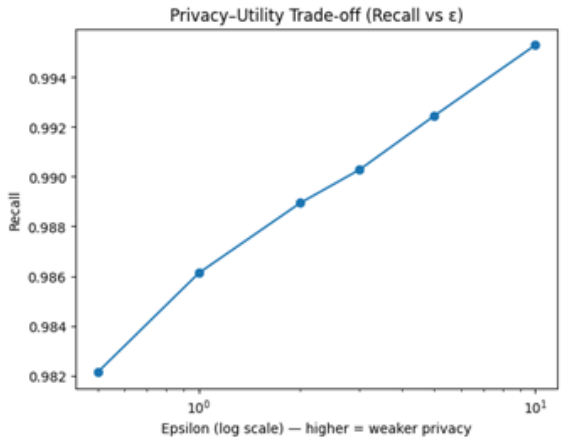

Figure 4.4: Recall of the DP-MLP model under different privacy budgets (ϵ). Higher ϵ values correspond to weaker privacy constraints and improved detection performance

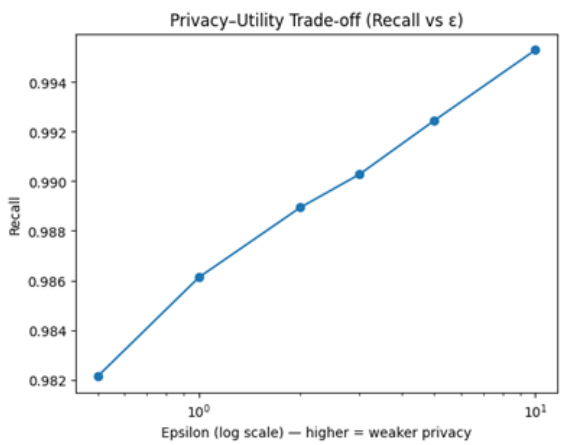

Figure 4.5: F1-score of the DP-MLP model under different privacy budgets (ϵ), illustrating the privacy–utility trade-off

Figures 4.4 and 4.5 illustrate the privacy–utility trade-off. As ϵ increases, both Recall and F1-score steadily improve. For example, Recall rises from 98.21% at $\epsilon = 0.5$ to 99.53% at $\epsilon = 10$, while the F1-score increases from 98.54% to 99.38%.

This is a expected behavior, because of the larger ϵ values correspond to weaker privacy constraints and reduced noise injection during training. Conversely, smaller ϵ values introduce stronger noise, which slightly decreases predictive performance.

	Accuracy	Precision	Recall	F1	AUC	Target_epsilon
0	0.985456	0.988705	0.982132	0.985408	0.998703	0.5
1	0.988103	0.990042	0.986125	0.988079	0.999056	1.0
2	0.990046	0.991135	0.988938	0.990035	0.999230	2.0
3	0.990952	0.991618	0.990275	0.990946	0.999312	3.0
4	0.992192	0.991950	0.992438	0.992194	0.999415	5.0
5	0.993775	0.992285	0.995287	0.993784	0.999518	10.0

	Achieved_epsilon
0	0.495620
1	0.998758
2	1.995702
3	2.993148
4	4.996963
5	9.999836

Table 4.1: Performance metrics of the DP-MLP model for different privacy budgets (ϵ)

This stable, monotonic pattern confirms our hypothesis regarding the theoretical relationship between privacy strength and model utility.

The detailed numerical results for each privacy level are summarized in Table 4.1.

4.5 Comparative Benchmark Results

The baseline MLP, DP-MLP and Random Forest were compared on the same test set directly employing standard classification metrics. The aggregated results are reported in Table 4.2

	Accuracy	Precision	Recall	F1	AUC	TP
Baseline MLP	0.998224	0.999137	0.997309	0.998222	0.999951	56710
DP-MLP	0.993933	0.992616	0.995269	0.993941	0.999465	56594
Random Forest	0.993915	0.998970	0.988850	0.993884	0.999879	56229

	TN	FP	FN
Baseline MLP	56814	49	153
DP-MLP	56442	421	269
Random Forest	56805	58	634

Table 4.2: Performance comparison of the baseline MLP, DP-MLP, and Random Forest models on the test set

The baseline MLP produces the best overall performance of 99.92% accuracy and AUC 0.99951. DP-MLP also shows a similar accuracy and AUC of 99.39% and 0.99947 compared to the baseline, showing only a small reduction despite the added privacy constraints. The Random Forest model performs slightly worse in

terms of recall (98.88%), meaning that more fraudulent transactions are missed than neural models. On the whole, our differentially private model is competitive against the baseline method showing that with little loss in predictive ability, privacy preservation is feasible within a reasonable framework.

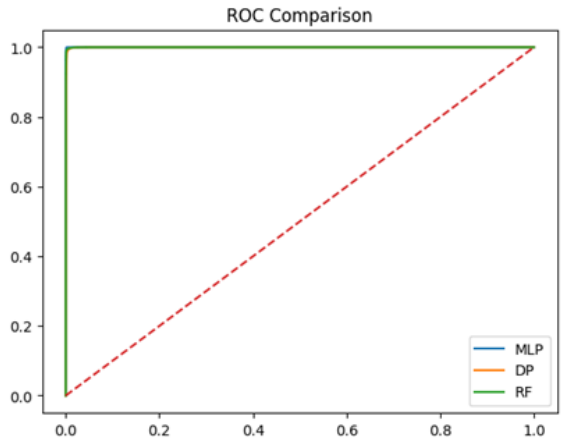

Figure 4.6: ROC curves of the baseline MLP, DP-MLP, and Random Forest models, showing high discrimination performance for all approaches

Figure 4.6 presents the ROC curves of all evaluated models. All curves are concentrated near the upper left corner, indicating strong discrimination capability. The baseline MLP achieves the highest AUC, followed closely by the DP-MLP, while the Random Forest shows slightly lower performance. The small gap between the baseline and DP-MLP curves further confirms that incorporating Differential Privacy does not significantly degrade predictive accuracy.

4.6 Blockchain Log Outputs

Besides predictive performance, the integrity of the training process was tested with a blockchain inspired logging mechanism. While conducting the experiments, each model's performance metrics were stored as hash linked entries, forming an immutable chain.

To verify tamper resistance, an integrity check was conducted. This chain was validated successfully at first. Once a single value had been manually modified, the process identified the inconsistency and subsequently identified the chain as invalid. The observed outputs were:

Chain valid: True After tampering: False

This confirms that any modification to previously recorded entries immediately breaks the hash chain and can be detected automatically.

The results indicate that any unauthorized change in the recorded training metrics would break the hash linkage immediately and be instantly detectable. Thus,

the proposed logging mechanism increases transparency, reproducibility, and the overall trustworthiness of the fraud detection pipeline.

4.7 Discussion

This research analyzed the performance of privacy-preserving machine learning method with the blockchain based logging system for fraud detection.All models have reliable accuracy as exhibited by the experimental results, although the baseline MLP shows the best performance, followed closely in the following phase by the DP-MLP by little decrease. This means that adding Differential Privacy does not greatly negatively affect classification quality.

The comparison between models also shows the good property of neural networks for structured fraud detection tasks. Both baseline MLP and DP-MLP have consistent recall superiority in comparison to Random Forest, possibly due to deep learning models capturing more complex nonlinear relationships in transaction data. Yet the Random Forest is still a robust and interpretable classical benchmark.

The privacy utility analysis further confirms the expected trade-off between privacy strength and model performance. Lower privacy budgets introduce more noise and slightly reduce predictive metrics, while higher budgets are more accurate and produce better recall.However, even under stronger privacy constraints, the DP-MLP maintains competitive performance, demonstrating that meaningful privacy protection can be achieved without incurring a noticable loss of utility.

The blockchain based logging mechanism not only improves the ability to predict but also provides greater integrity and transparency to the training process. Any unauthorized changes (such as by storing data in a hash linked format) of training metrics can be immediately identified.

The findings in general indicate that integrating Differential Privacy with secure logging presents the combination of both to find a feasible and efficient approach for user friendly, yet privacy conscious fraud detection. Both data confidentiality and process integrity and high predictive performance are still achieved if the proposed framework is implemented, with the data integrity of the data.

Chapter 5 Extended Application Scenarios ERP and Limitations

This chapter discusses how the proposed framework can be applied in real life environments. Explores the potential integration of the system within ERP infrastructures and highlights practical considerations such as scalability, deployment challenges, and operational constraints.

5.1 Future Work

The experimental results show that adding privacy preserving learning and blockchain based integrity mechanisms does not noticeably reduce the model performance. All models still achieve high recall, F1-score, and AUC values, and the effect of Differential Privacy is small.

Although this thesis mainly focuses on fraud detection, the framework can also be used in other application areas. Privacy preserving learning and tamper resistant logging may be useful in domains where sensitive data must be protected and where transparency and accountability are important, such as healthcare, finance, supply chain management, and large enterprise systems.The integration of blockchain and machine learning has been identified as an important research direction, as secure and efficient data sharing remains a key challenge in practical systemsChen et al. (2020)

On the other hand, the current work uses Differential Privacy together with a blockchain based logging approach to protect data and to monitor the training process.The lightweight hash chained logger produces tamper evident records at very low computational cost, but it is not a fully decentralized or distributed blockchain system. For this reason, future studies can explore permissioned or enterprise blockchain platforms and test the scalability, fault tolerance, and real time performance in practical environments. Research on these aspects could be useful

to enhance transparency and trust when the system is put into action in actual applications.

In addition, the present experiments were conducted on a single dataset under controlled laboratory conditions. Future studies can evaluate the framework on larger and more diverse real world financial datasets to better assess robustness and generalization. Testing the system on highly imbalanced or noisy transaction data would provide a more realistic fraud detection scenario.

Furthermore, other privacy preserving approaches may be explored. Besides Differential Privacy, techniques such as federated learning or secure multi-party computation could be integrated to further enhance confidentiality, especially when sensitive data cannot be centrally stored.

Recent studies also indicate that the integration of blockchain technologies with enterprise systems remains an emerging area with limited practical implementations and open research challenges. Therefore, extending the proposed framework beyond experimental settings and adapting it to real enterprise infrastructures represents an important next step.

Considering these directions, the following section discusses how the proposed privacy preserving fraud detection framework can be integrated into Enterprise Resource Planning (ERP) systems for practical and real world use.

5.2 ERP Integration Potential

Of these domains, Enterprise Resource Planning (ERP) systems represent a particularly relevant and practical environment.ERP platforms are widely used to automate and manage core business processes such as finance, accounting, procurement, and logistics. Integrating blockchain with ERP systems offers important advantages such as decentralization, increased transparency, and improved traceability of business records while maintaining existing enterprise workflows Kitsantas (2022).

On a governance and compliance level, blockchain–ERP integration can help improve data security and process integrity. In particular, blockchain-based immutable ledger or "data vault" approaches make it possible to store transactions from finance and accounting in tamper-resistant forms that provide greater auditability and protection against unauthorized modifications Dalal & Abdul (2017).

Previous studies have also highlights the effectiveness of blockchain technologies in supply chain management, which is a core ERP function. Blockchain enabled traceability solutions enhance transparency, trust, and information sharing among various stakeholders (specifically in sensitive sectors i.e. food, pharmaceuticals, logistics). Dalal & Abdul (2017) These findings indicate that blockchain is not only

useful for fraud detection but also beneficial for broader enterprise processes that require verifiable and trustworthy data sharing.

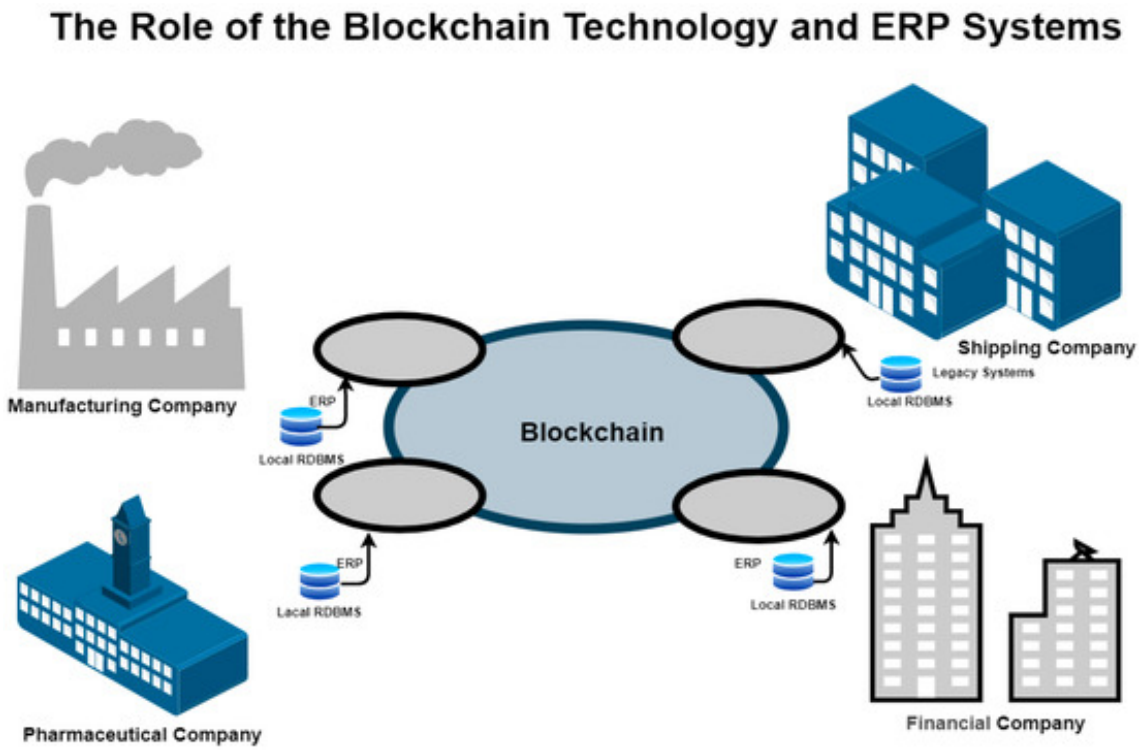

Figure 5.1: Integration of ERP system with Blockchain

Despite these benefits, several practical challenges must be considered. The integration of blockchain solutions in ERP environments can result in increasing latency, scalability and storage constraints. The processing speed, and the overhead of the network are problematic for the system and can potentially degrade system flexibility as the transactional volume grows Sunmola & Lawrence (2024). Therefore, real-world implementations must carefully balance security improvements with operational efficiency.

The findings of this thesis also show, from an enterprise system perspective, how privacy sensitive analytics can be integrated into the ERP financial module without loss of predictive value. DP-SGD allows us to train for the existing models with no individual level information being revealed, and an inductive logging mechanism in the form of a blockchain maintains an immutable history of training and evaluation records. All together, these elements have been shown to improve not only confidentiality but also integrity of ERP workflows

Figure 5.2 illustrates a proposed integrated ERP architecture demonstrating how a privacy-preserving fraud detection model and blockchain based logging can operate within a typical enterprise process pipeline.

The transactional data created by the ERP finance module is then sent to the

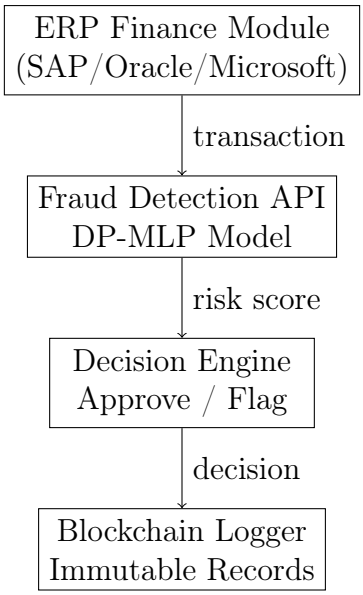

Figure 5.2: Proposed integrated ERP privacy preserving fraud detection architecture

fraud detection API, and the DP-MLP model assesses the risk level of each transaction first. A decision engine is activated from the above model, either approving or flagging the transaction for further inspection. Lastly, all model outputs and decisions are logged in the blockchain logger for tamper-proof and immutable audit trails. This architecture allows for both secure analytics and transparent monitoring within enterprise workflows.

Overall, the findings indicate that the proposed framework is applicable not only to fraud detection but also to a wider range of enterprise scenarios where secure, privacy-aware, and auditable analytics are required.

5.3 Practical Considerations

The integration of blockchain technologies into Enterprise Resource Planning (ERP) environments involves both technical and organizational challenges. Systematic analyses indicate that successful adoption depends not only on technological readiness but also on organizational alignment, process redesign, and regulatory considerations Sunmola & Lawrence (2024).

Although blockchain can enhance transparency, traceability, and integrity of enterprise records, its incorporation may introduce additional architectural complexity and implementation overhead. Studies emphasize that integrating distributed ledger technologies with existing ERP infrastructures requires careful planning to avoid disruption of established workflows Kitsantas (2022).

Moreover, blockchain-based solutions offer special opportunities in the fields of supply chain and multi-stakeholder settings, where immutable and verifiable records

lead to trust, accountability, and increased data sharing among participants Dasaklis et al. (2022). These features are relevant for ERP processes with multiple departments and partners.But, some limitations still exist. Several papers have consistently described scalability limitations, latency and storage costs as major constraints to big scale blockchain implementations. As transaction volume increases, these can adversely affect throughput and overall responsiveness of the system, as well as limit this capability practical feasibility for the high demand enterprise systems Sunmola & Lawrence (2024).

Therefore, organizations need to carefully balance the security and auditability benefits of blockchain technologies with operational performance requirements and infrastructure costs in practical ERP use cases.

Chapter 6 Conclusions

This final chapter summarizes the overall findings of this thesis and highlights the main outcomes of the proposed framework. The study investigated how fraud detection models can be combined with privacy preserving techniques and blockchain inspired logging to protect sensitive data while maintaining high predictive performance.

The following sections present the main contributions of the study, its limitations, and the final conclusions.

6.1 Contributions

This paper contributes as follows:

• A privacy preserving fraud detection method that contains a Multi-Layer Perceptron (MLP), Differential Privacy (DP-SGD), and a blockchain inspired logging mechanism in a single pipeline.

• An empirical research about the privacy–utility trade-off by training DP-MLP models with several privacy budgets and investigating their effect on the accuracy, recall and F1-score.

• The design and implementation of a lightweight hash chained logging structure that provides tamper evident and transparent training records without requiring a full blockchain infrastructure.

• A comparative benchmark on the same dataset between classical machine learning (Random Forest), standard neural networks, and differentially private neural networks.

• A practical integration scenario demonstrating how the proposed framework can be used in real world industries, and how can adapt in ERP based financial systems.

6.2 Limitations

Despite the promising results, several limitations should be acknowledged. First, the dataset used in this study has relatively strong class separability and a balanced distribution. This makes the classification task easier compared to many real life fraud detection, where data is often highly imbalanced and noisy. For this reason, the near-perfect performance observed in the experiments may not fully represent the difficulties faced in enterprise environments.

Second, the experiments were conducted in an offline evaluation setting. In practical ERP deployments, additional constraints such as latency, scalability, and real-time processing requirements become important. Therefore, the framework should be tested in a more realistic setup to understand how it behaves under operational workloads (e.g., throughput and response time) Sunmola & Lawrence (2024).

Third, the blockchain-inspired logging mechanism is implemented as a lightweight hash chained structure. It provides tamper evident records, but it is not a fully distributed blockchain network. Future work may integrate permissioned or enterprise blockchain platforms to study decentralization, governance, and fault tolerance in real deployments Kitsantas (2022).

Finally, while Differential Privacy is applied in this thesis, other privacy-preserving approaches (such as federated learning or secure multi-party computation) could be explored as complementary directions, especially when data cannot be centralized.

6.3 Conclusions

This thesis presented a privacy preserving and integrity-aware fraud detection framework that can be extended to enterprise resource planning (ERP) environments. The proposed approach combines a neural network-based classifier, Differential Privacy using DP-SGD, and a blockchain-inspired logging mechanism to support sensitive financial data processing.

The experimental results show that the baseline neural network achieves very high predictive performance, and the differentially private model maintains similar performance with only a small decrease. The privacy–utility trade-off analysis also indicates that stronger privacy settings generally reduce performance slightly, while weaker privacy settings improve it, which is consistent with the expected behavior.

In addition, the blockchain-inspired logging component provides a practical way to create tamper-evident audit records during training and evaluation. This improves transparency and supports traceability of model-related outputs without requiring a full blockchain deployment.

Overall, the study shows that privacy protection, process integrity, and strong predictive performance can coexist within one machine learning pipeline. By connecting privacy-preserving learning with traceable logging, this work supports the development of more secure and trustworthy analytics that can be considered for real-world enterprise applications.

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